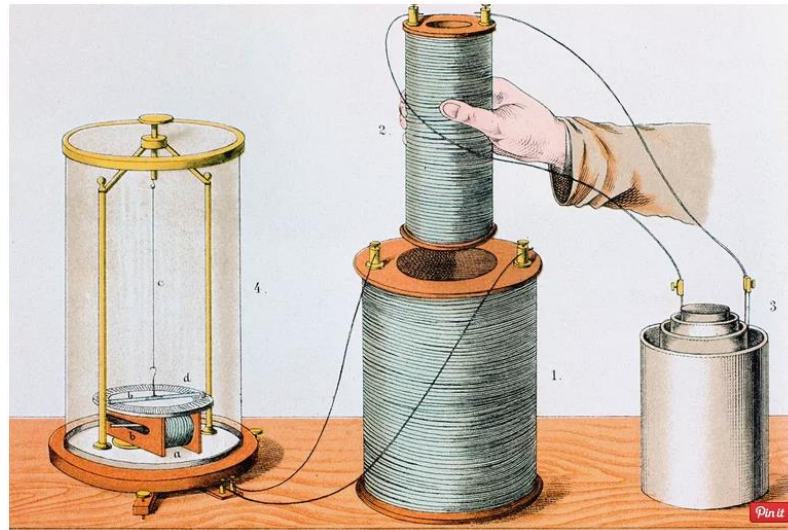


Chapter 29

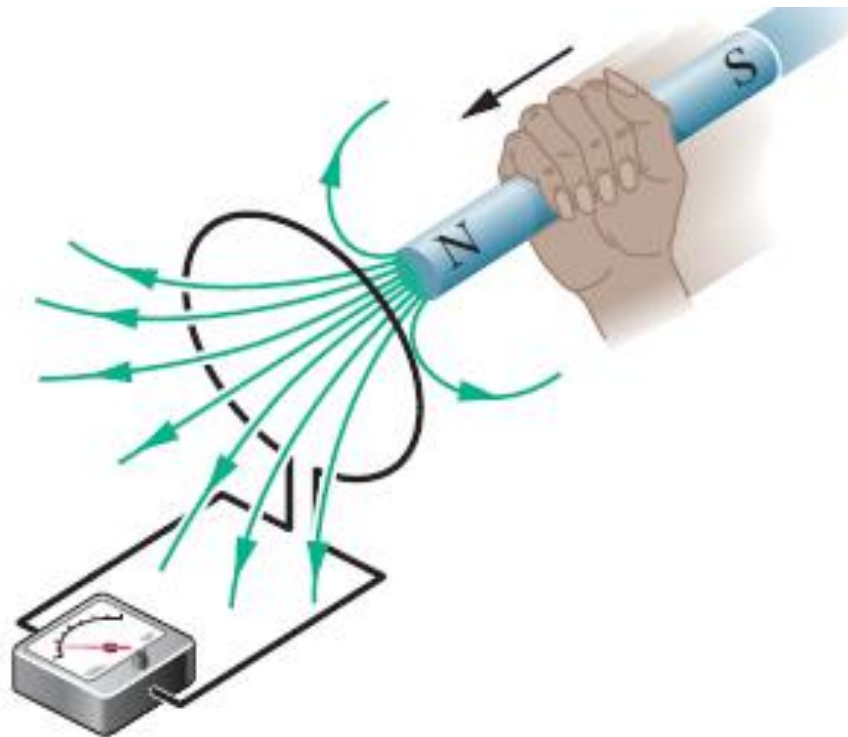
Electromagnetic Induction



Induction Experiments

Faraday's experiments: *Induced currents.*

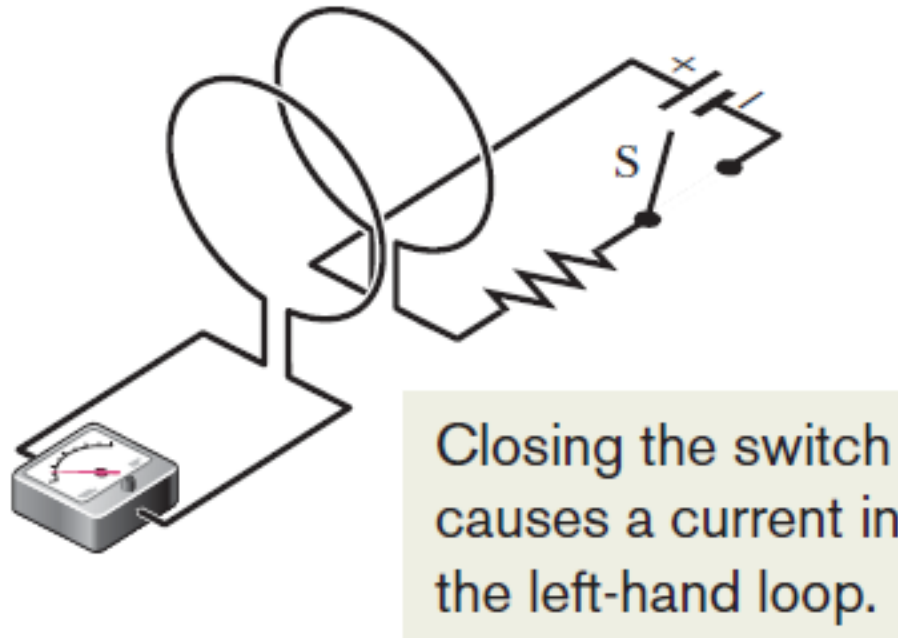
- An ammeter registers a current in the coil when the magnet is moving with respect to the loop.



Induction Experiments

Faraday's experiments: *Induced currents.*

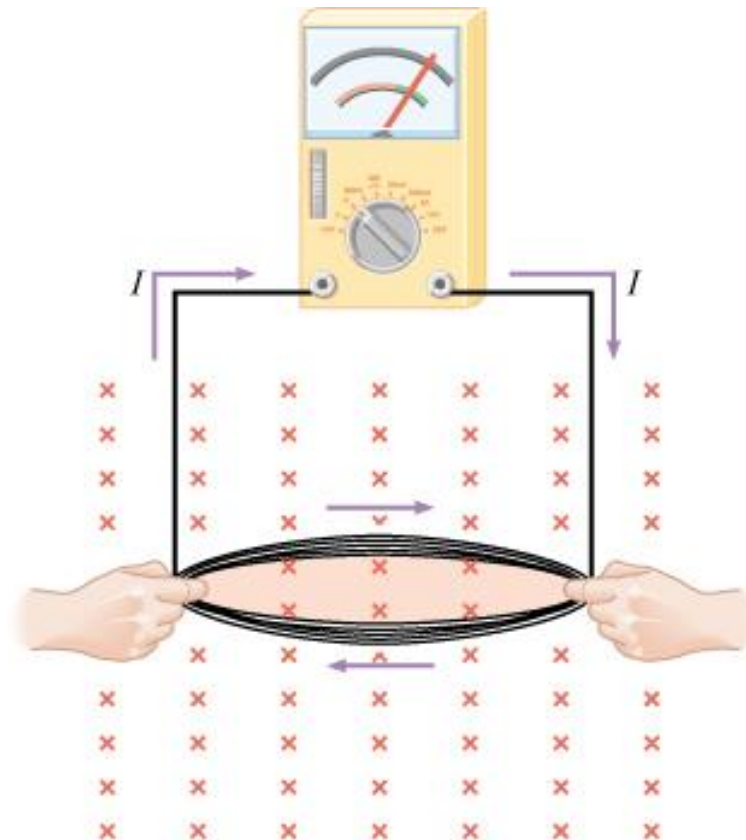
- An ammeter registers a current in the left just as the switch S is closed (or opened) in the right coil. No motion of the coils is involved.



Induction Experiments

Faraday's experiments: *Induced currents.*

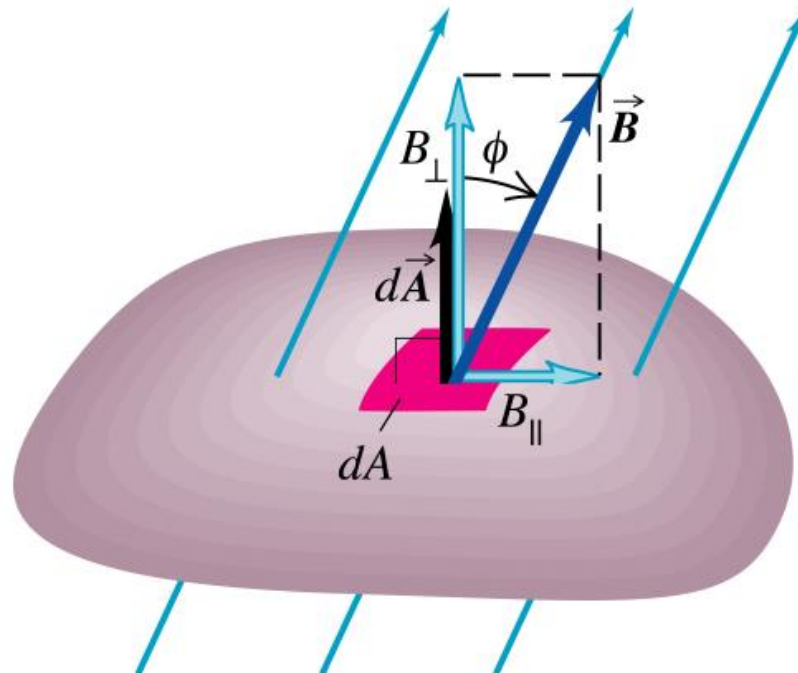
- A current be induced by changing the *area* of a coil in a constant magnetic field.



Faraday's Law

The common element in these experiments is the **changing** magnetic flux through a circuit.

$$\Phi_m = \iint B_{\perp} dA = \iint \vec{B} \cdot d\vec{A}$$



Faraday's Law

In each experiment, both a *voltage* and a *current* are induced because the coil forms a complete circuit. If the circuit were open, there would be no induced current, but there would be an induced voltage – an induced emf $\mathcal{E}$.

Faraday's Law of Induction:

The induced emf in a closed loop equals the negative of the time rate of change of the magnetic flux.

$$\mathcal{E} = -\frac{d\Phi_m}{dt}$$

*The negative (–) sign is explained by **Lenz's Law**.*

Faraday's Law

Changing magnetic flux

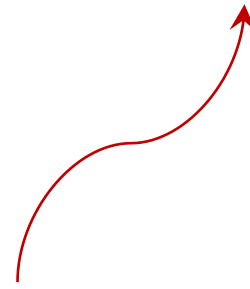
$$\Phi_m = B_{\perp} dA = BA \cos \phi$$

$$\frac{d\Phi_m}{dt} = (A \cos \phi) \frac{dB}{dt} + (B \cos \phi) \frac{dA}{dt} + (-BA \sin \phi) \frac{d\phi}{dt}$$

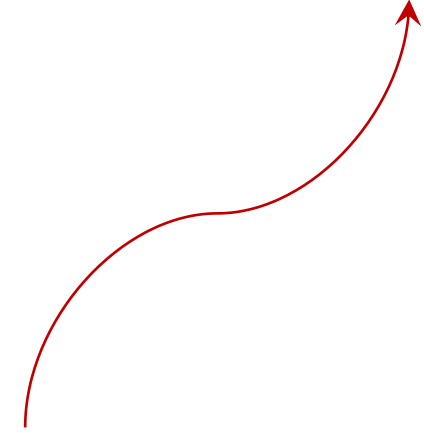
*Changing magnetic
field strength*



Changing area



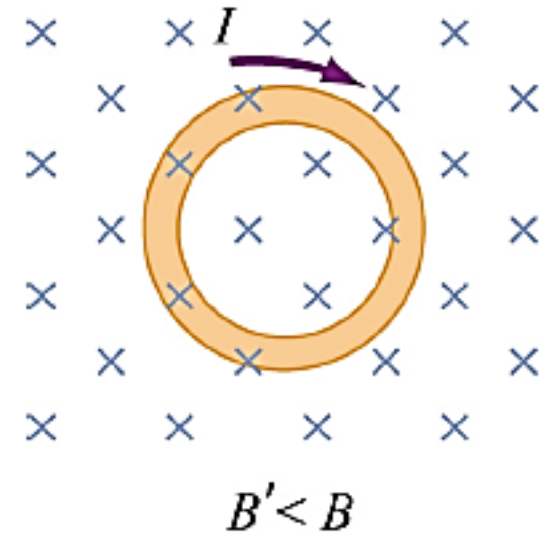
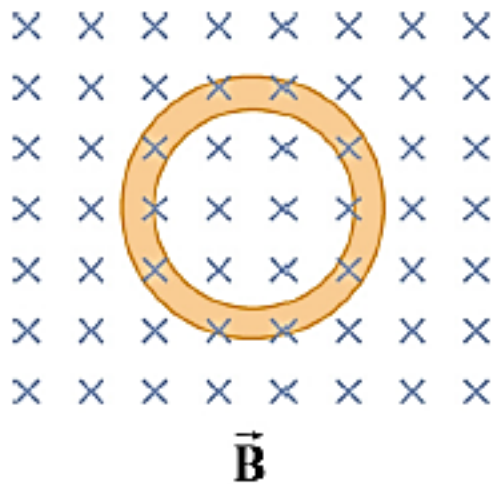
Changing orientation



Faraday's Law

Changing magnetic field strength

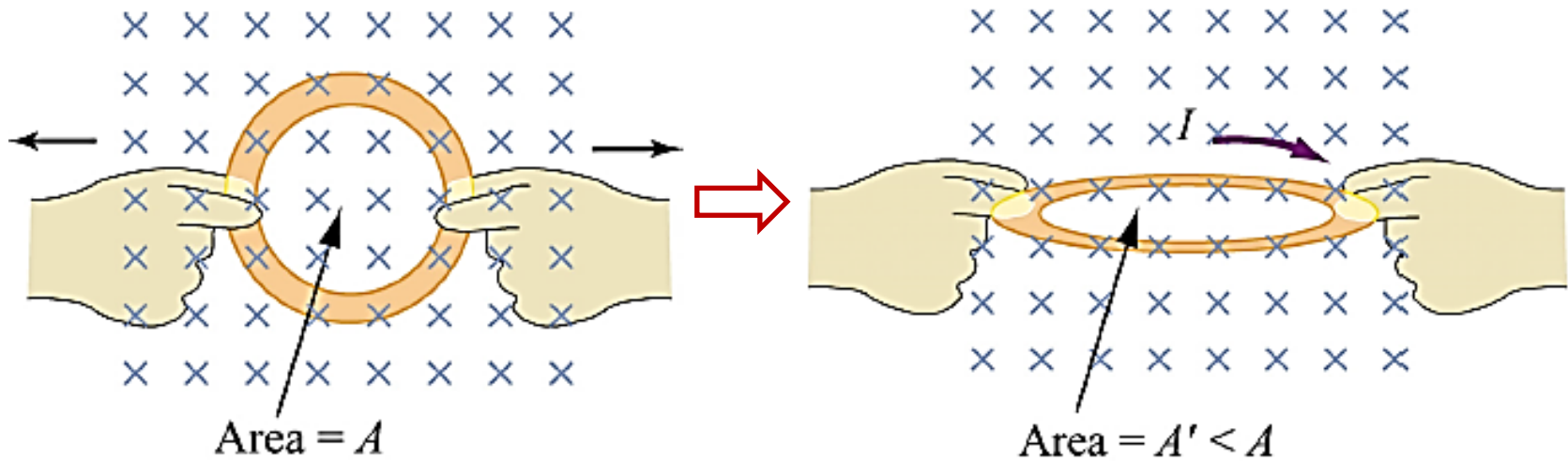
$$\frac{d\Phi_m}{dt} = (A \cos \phi) \frac{dB}{dt}$$



Faraday's Law

Changing loop area

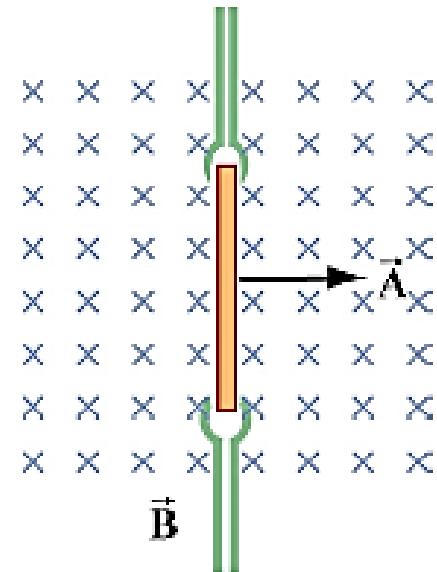
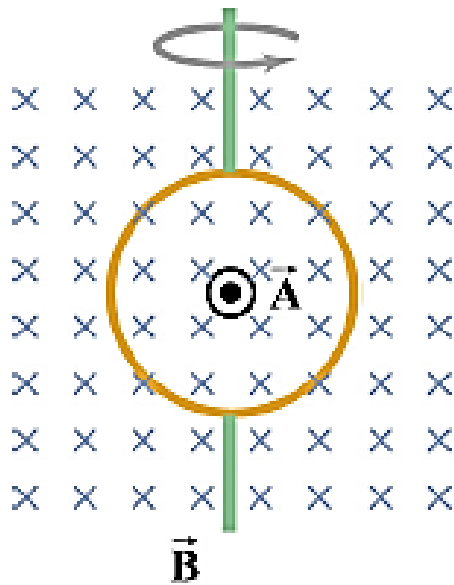
$$\frac{d\Phi_m}{dt} = (B\cos\phi) \frac{dA}{dt}$$



Faraday's Law

Changing loop area

$$\frac{d\Phi_m}{dt} = (-BA\sin\phi) \frac{d\phi}{dt}$$



Faraday's Law

To determine the **direction** of the induced current :

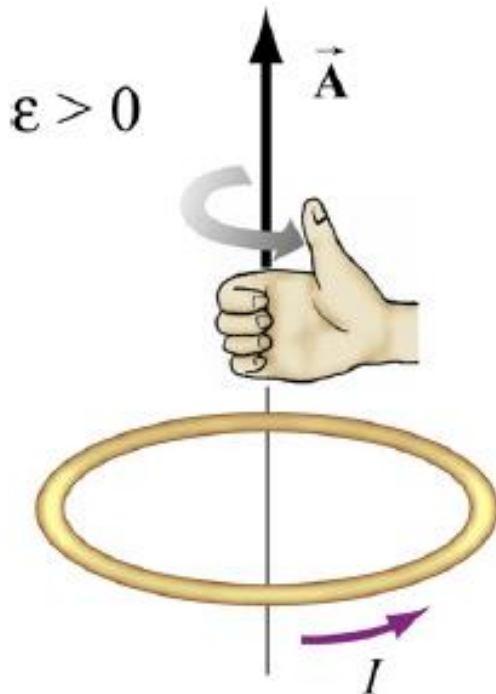
1. Define a positive direction for the area vector $\vec{A}$.
2. Assuming that $\vec{B}$ is uniform, take the dot product of $\vec{B} \cdot \vec{A}$. This allows for the determination of the sign of the magnetic flux Φ_B .
3. Obtain the rate of flux change $d\Phi_B/dt$ by differentiation. There are three possibilities:

$$\frac{d\Phi_B}{dt} \begin{cases} > 0 \Rightarrow \mathcal{E} < 0 \\ < 0 \Rightarrow \mathcal{E} > 0 \\ = 0 \Rightarrow \mathcal{E} = 0 \end{cases}$$

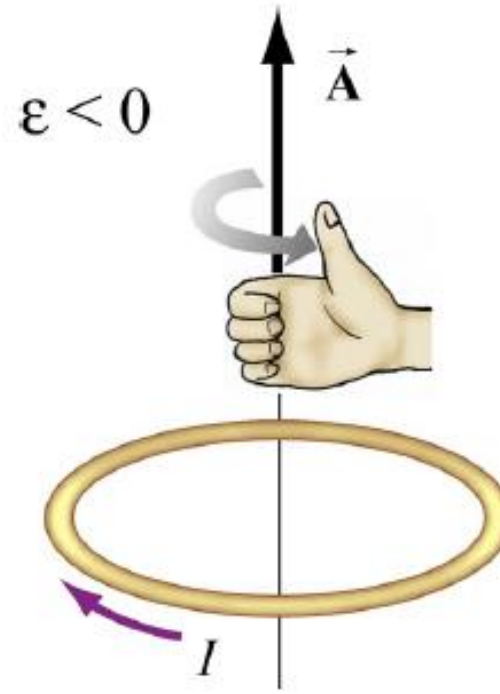
Faraday's Law

To determine the **direction** of the induced current (cont'd):

4. Determine the direction of the induced current using the *right-hand rule*.
With your thumb pointing in the direction of $\vec{A}$, curl the fingers around the closed loop.



Positive current

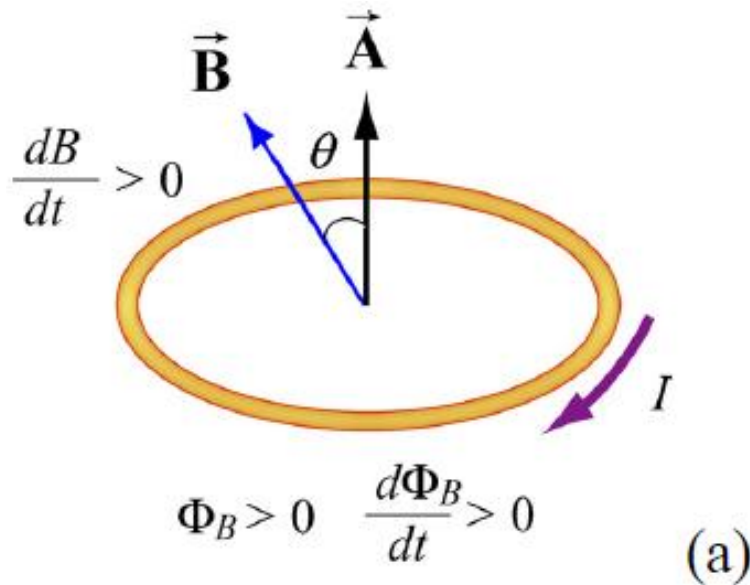


Negative current

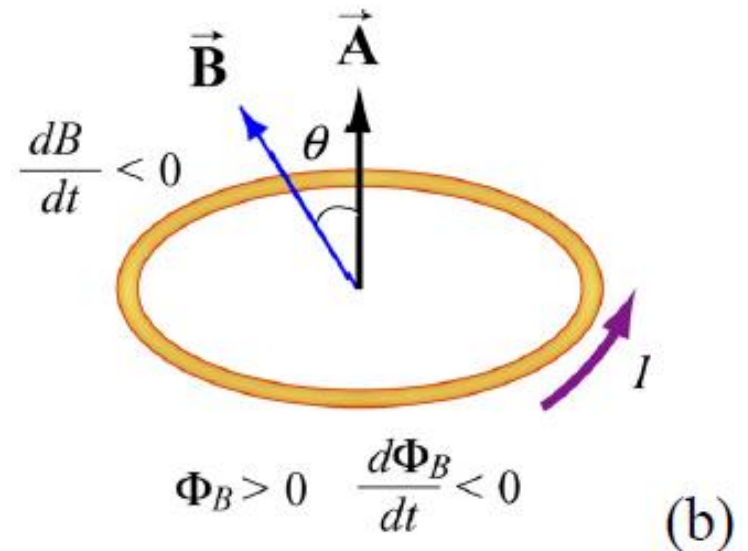
Faraday's Law

To determine the **direction** of the induced current (cont'd):

- Example



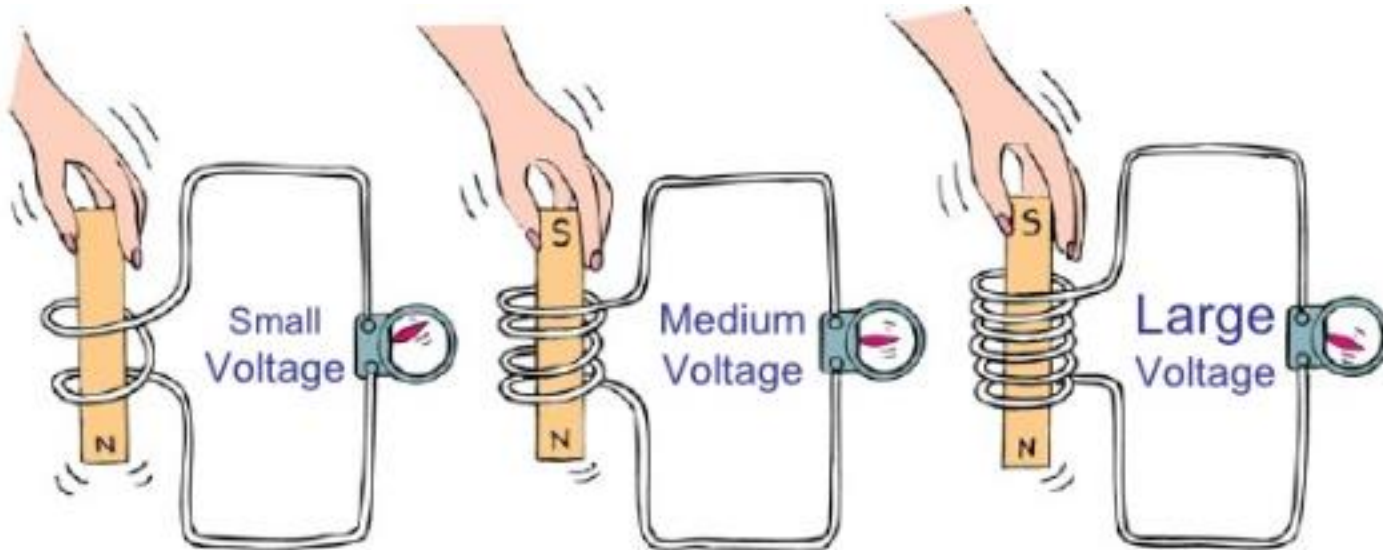
*Increasing flux
Negative current*



*Decreasing flux
Positive current*

Faraday's Law

For a coil with N identical turns:



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- *Total induced emf in coil with N turns*

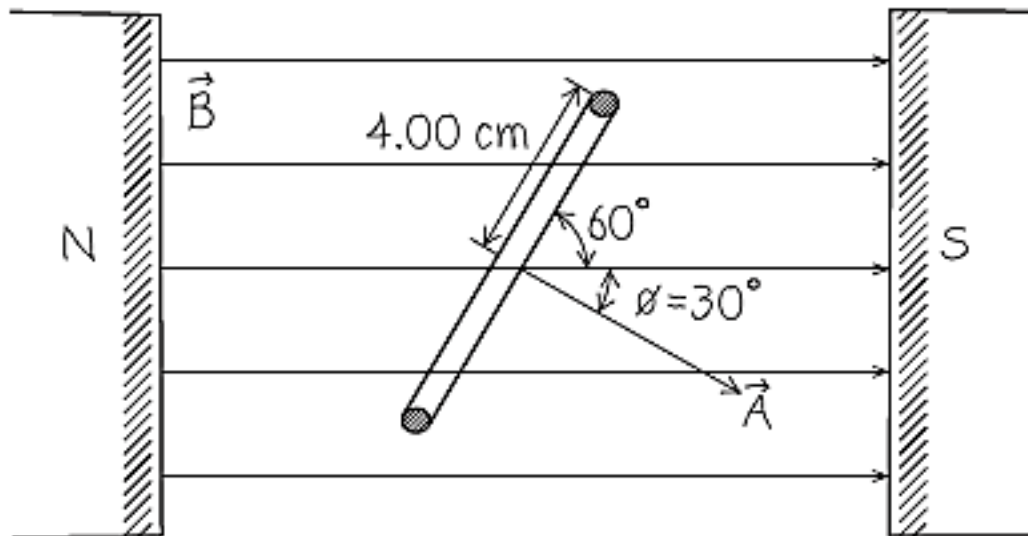
$$\mathcal{E} = -N \frac{d\Phi_m}{dt}$$

Faraday's Law

Example: Changing B -field

A coil of wire with 500 circular loops with radius 4.00 cm is placed between the poles of a large electromagnet, where the electric field is uniform and at an angle of 60° with the plane of the coil.

The field *decreases* at a rate of 0.20 T/s. What are the magnitude and direction of the induced emf?



Faraday's Law

Example: Changing B -field

Solution:

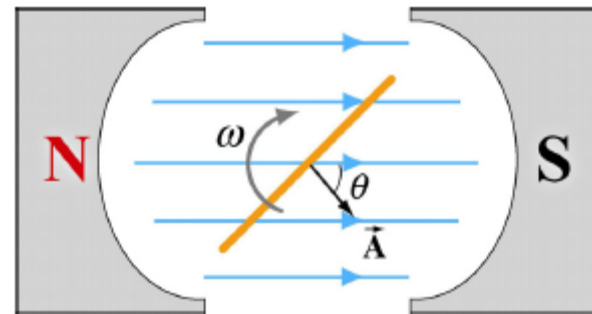
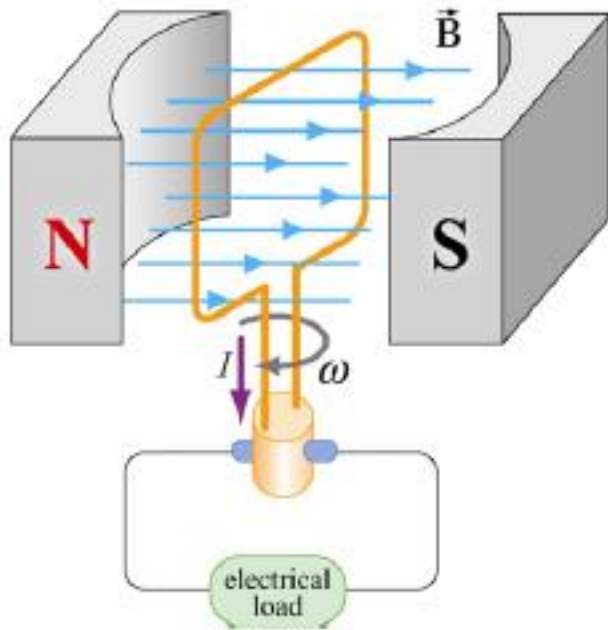
$$\begin{aligned}\frac{d\Phi_m}{dt} &= \frac{dB}{dt} A \cos 30^\circ \\ &= (-0.20 \text{ T/s})(0.00503 \text{ m}^2)(0.866) \\ &= -8.71 \times 10^{-4} \text{ T} \cdot \text{m}^2/\text{s}\end{aligned}$$

$$\begin{aligned}\mathcal{E} &= -N \frac{d\Phi_m}{dt} \\ &= -(500)(-8.71 \times 10^{-4} \text{ T} \cdot \text{m}^2/\text{s}) \\ &= 0.435 \text{ V (positive)}\end{aligned}$$

Faraday's Law

Example: A simple alternator

An N -turn rectangular loop is made to rotate with constant angular speed ω , as shown. The magnetic field is uniform and constant. At $t=0$, $\theta=0$. Determine the induced emf.



Faraday's Law

Example: A simple alternator

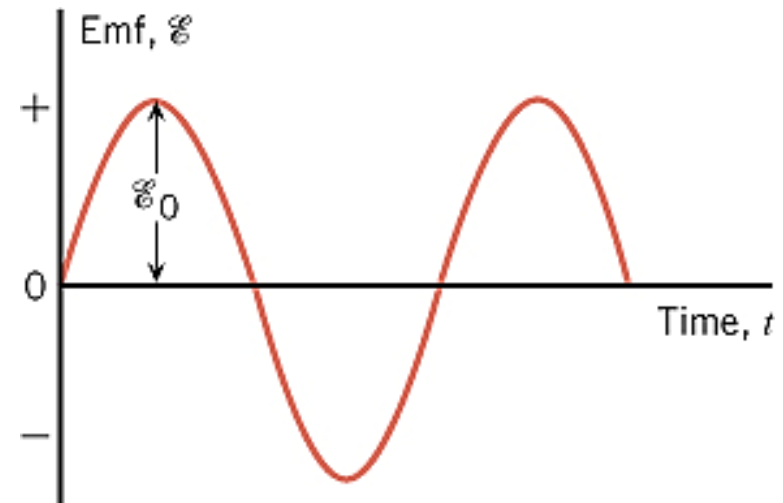
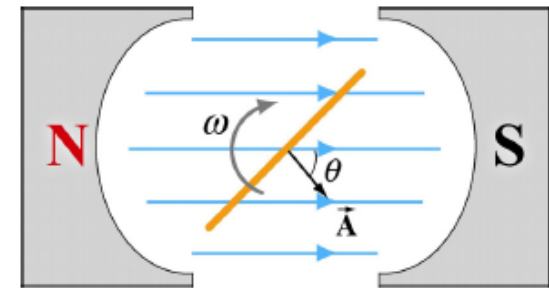
$$\frac{d\theta}{dt} = \omega \Rightarrow \theta = \omega t + \theta_0 = \omega t$$

$$\Phi_m = BA \cos(\omega t)$$

$$\frac{d\Phi_m}{dt} = -\omega BA \sin(\omega t)$$

$$\begin{aligned}\mathcal{E} &= -N \frac{d\Phi_m}{dt} \\ &= \omega NBA \sin(\omega t)\end{aligned}$$

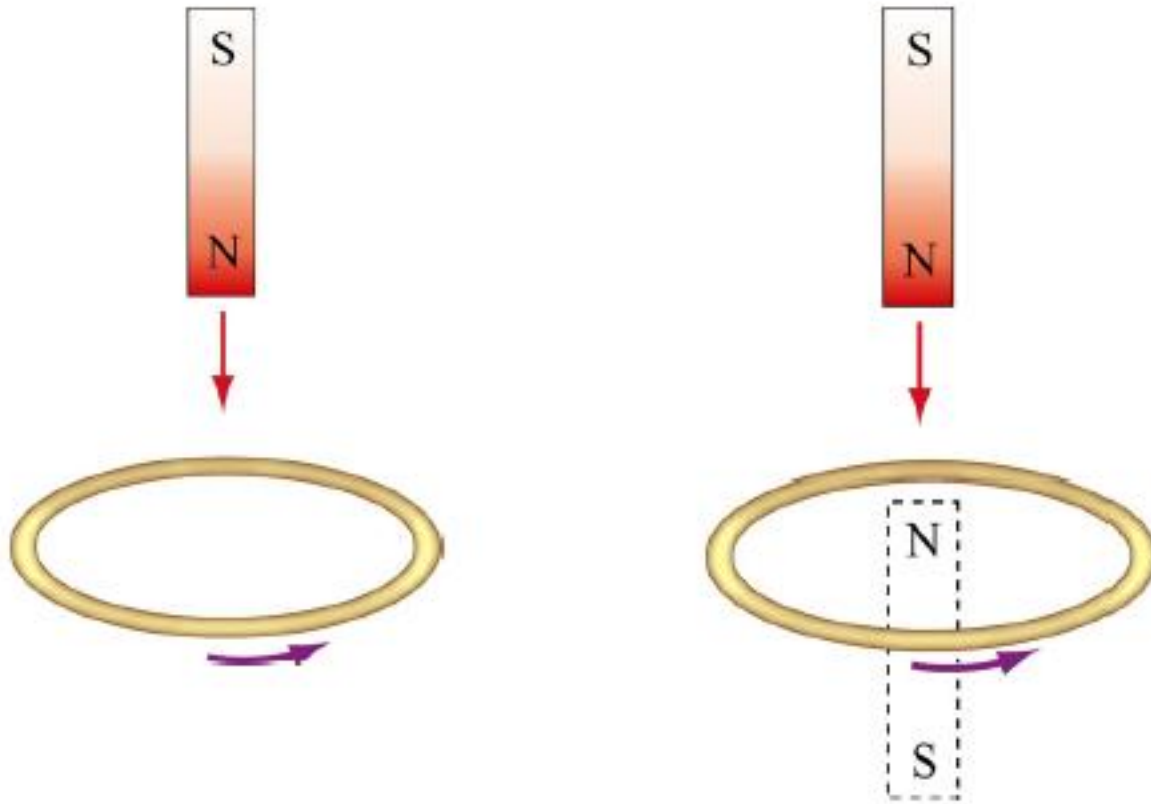
$$\mathcal{E} = \mathcal{E}_0 \sin(\omega t)$$



Lenz's Law

The direction of any magnetic induction effect is such as to oppose the cause of the effect.

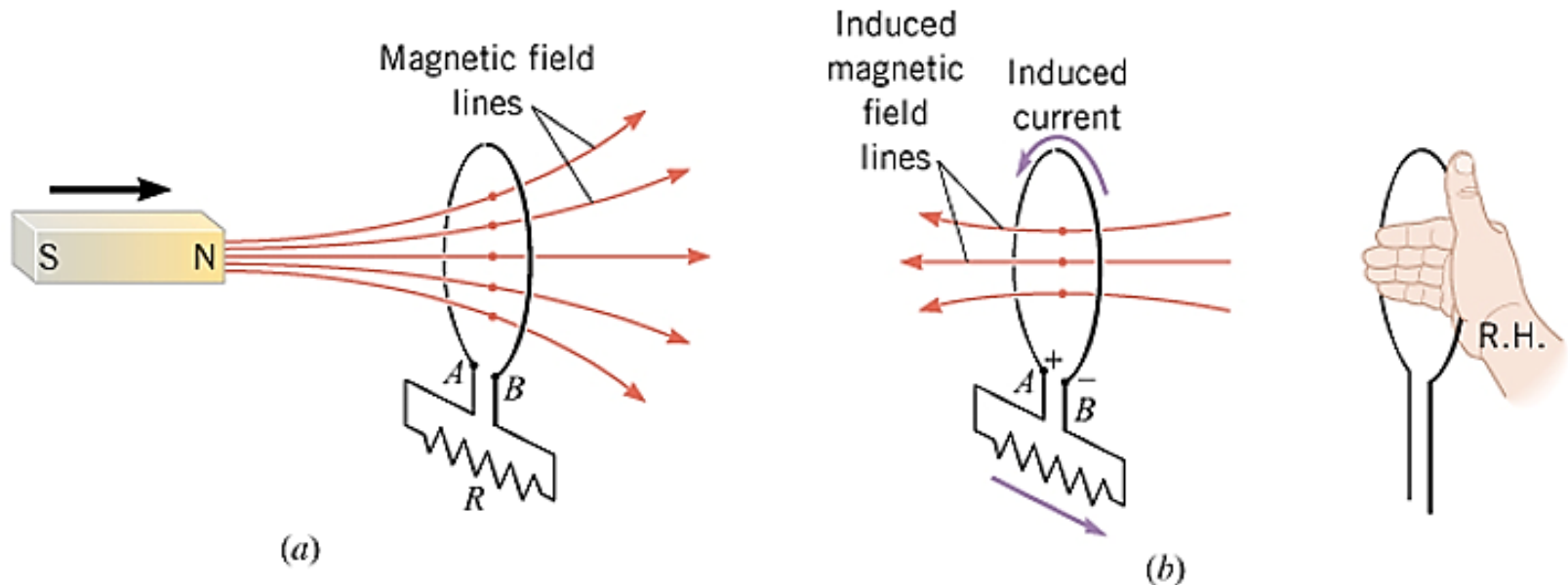
- When a bar magnet is thrust into a coil:



Lenz's Law

Direction of the induced current

1. Determine whether the magnetic flux that penetrates the coil is increasing or decreasing.
2. Find what the direction of the induced magnetic field must be so that it can oppose the change in flux by adding or subtracting from the original field.
3. Use RHR-2 to determine the direction of the induced current.

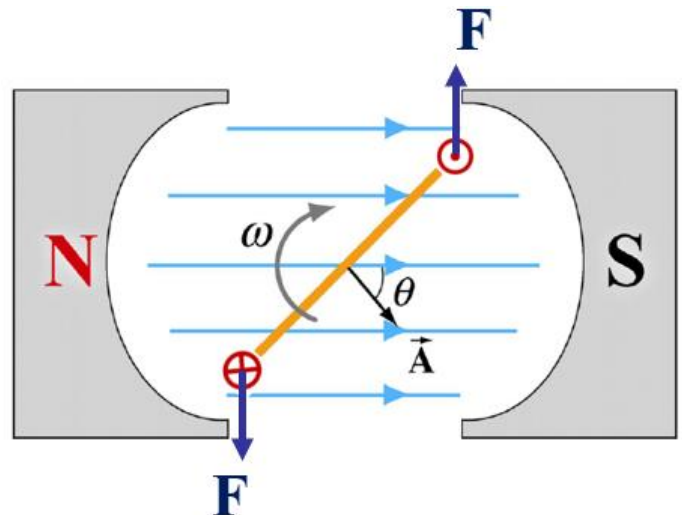
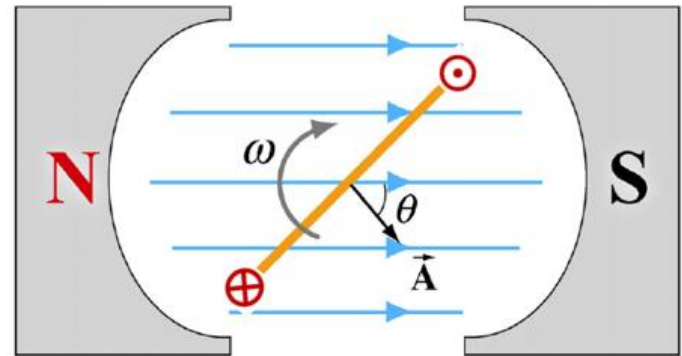


Lenz's Law

The direction of any magnetic induction effect is such as to oppose the cause of the effect.

Rotating loop of the alternator.

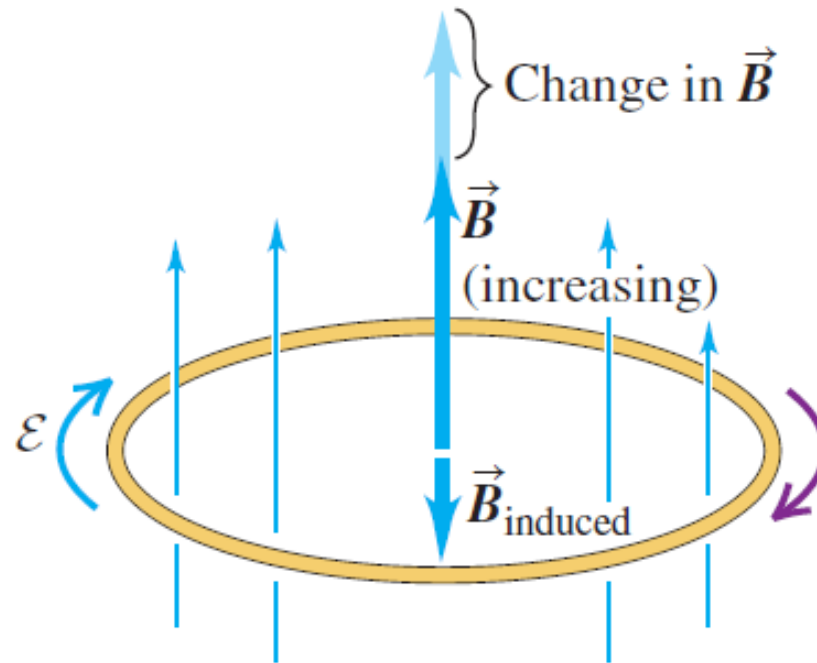
- a) The direction of the induced current.
- b) The induced current interacts with the magnetic field of the magnet. The force is in the direction shown.



Lenz's Law

Example: Lenz's law and the direction of induced current

There is a uniform magnetic field through the coil. The magnitude of the field is increasing, so there is an induced emf. Use Lenz's law to determine the direction of the resulting induced current.



Motional $\mathcal{E}$ mf

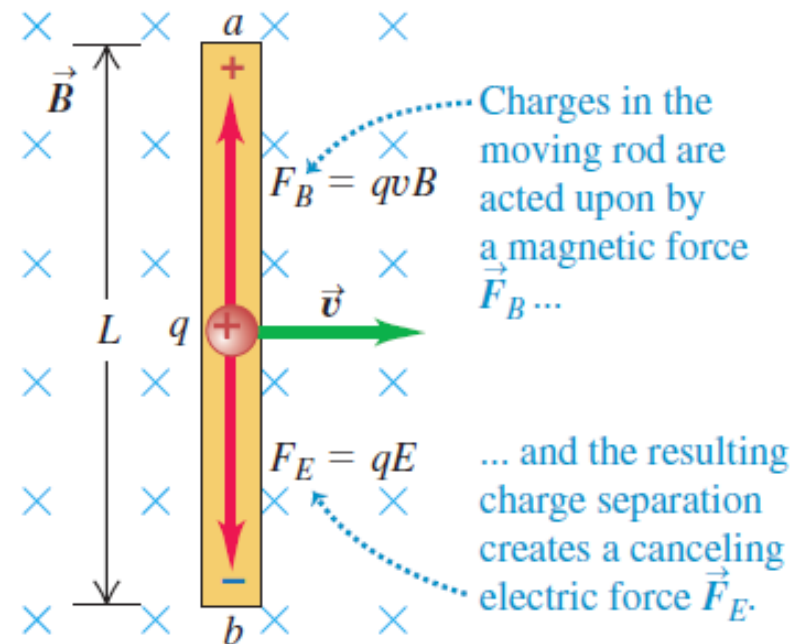
$\mathcal{E}$ mf induced in a moving conductor:

- Each charge within the conductor is moving and experiences a magnetic force $\vec{F}$

$$F = qvB$$

- The separation of charges result in an electric field $\vec{E}$ which, at steady state, balances the force

$$F = qvB = qE$$



Motional $\mathcal{E}$ mf

$\mathcal{E}$ mf induced in a moving conductor:

- The “induced” electric field $\vec{E}$

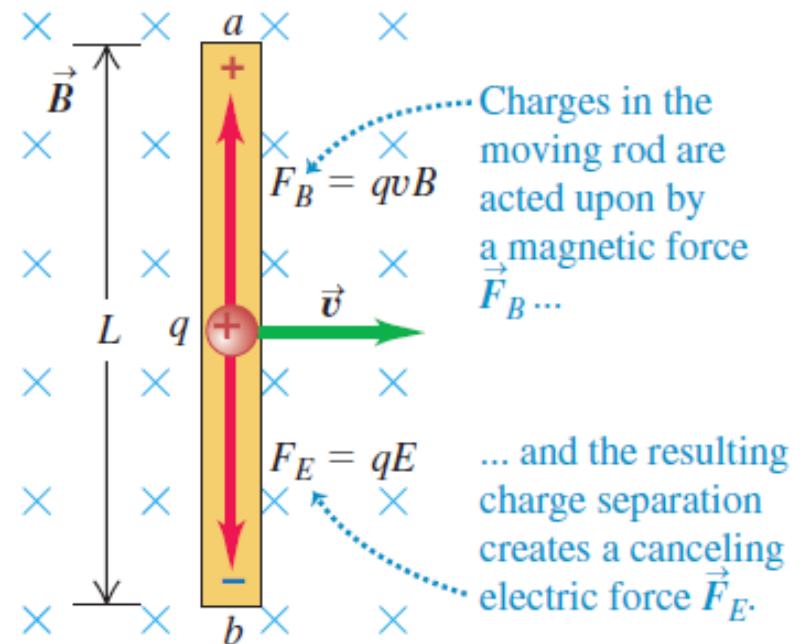
$$E = vB$$

- This is expressed in terms of an electric potential difference

$$V_+ - V_- = EL = vBL \equiv \mathcal{E}$$

- The induced $\mathcal{E}$ mf in the rod:

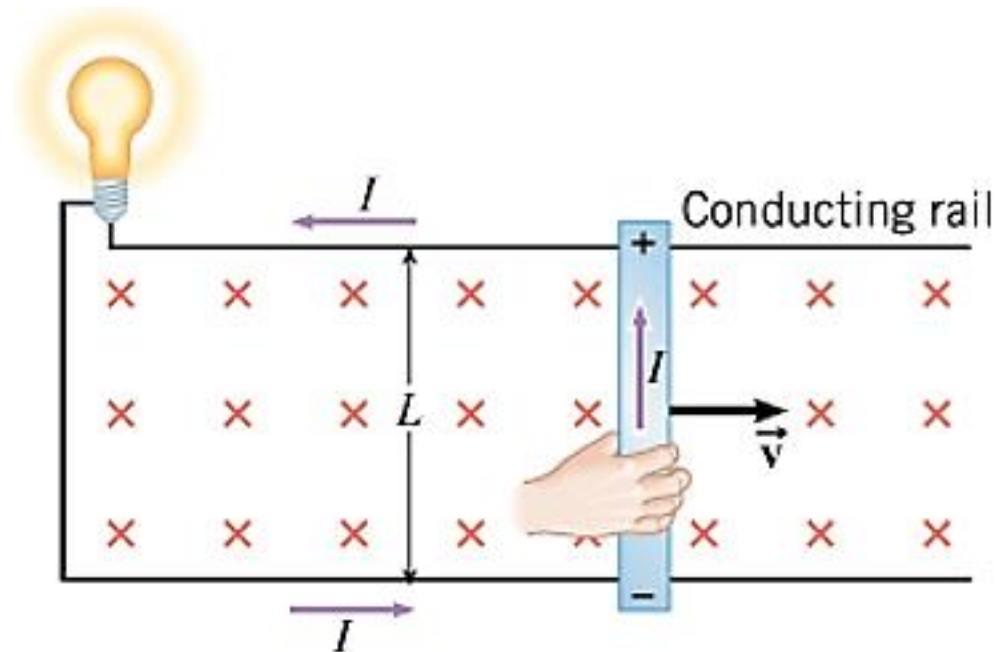
$$\mathcal{E} = vBL$$



Motional $\mathcal{E}$ mf

Example: Operating a Light Bulb with Motional $\mathcal{E}$ mf

Suppose the rod is moving with a speed of 5.0 m/s perpendicular to a 0.80 T magnetic field. The rod has a length of 1.6 m. The rod and rails have a negligible electrical resistance. The light bulb has a resistance of 96 Ω . Find a) the emf produced by the rod and; b) the current induced in the circuit.



Motional $\mathcal{E}$ mf

Example: Operating a Light Bulb with Motional $\mathcal{E}$ mf

Solution

a) the $\mathcal{E}$ mf produced by the rod.

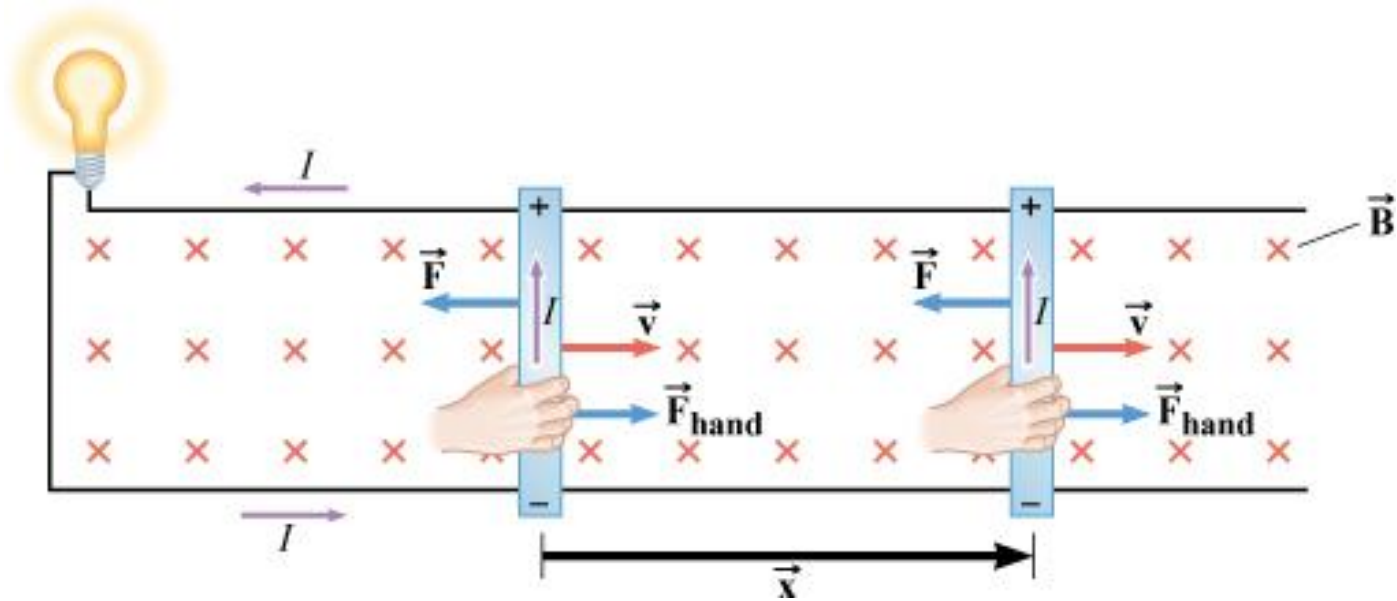
$$\mathcal{E} = vBL = (5.0 \text{ m/s})(0.80 \text{ T})(1.6 \text{ m}) = 6.4 \text{ V}$$

b) the current induced in the circuit.

$$I = \frac{\mathcal{E}}{R} = \frac{6.4 \text{ V}}{96 \Omega} = 0.067 \text{ A}$$

Motional $\mathcal{E}$ mf

Motional $\mathcal{E}$ mf and Electrical Energy.

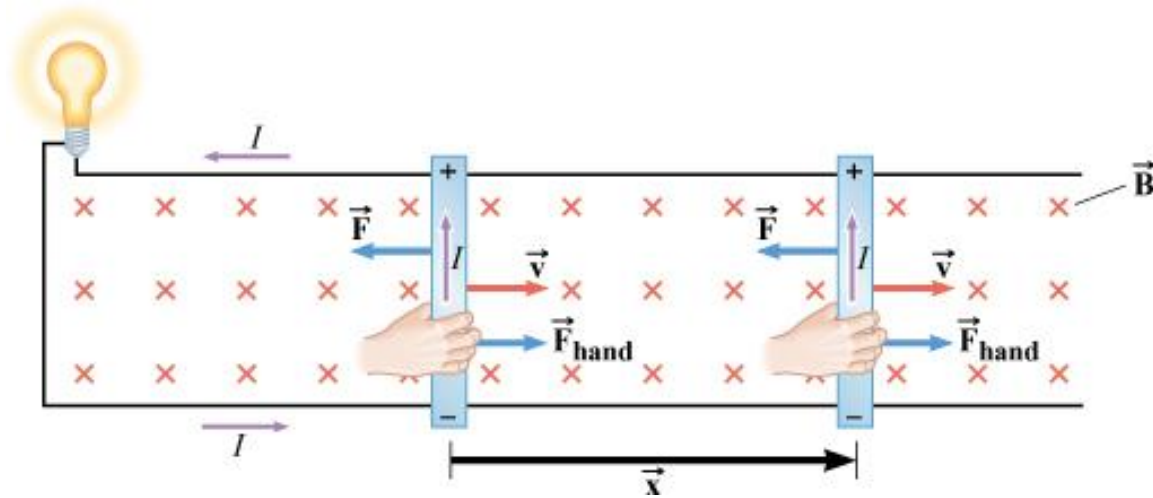


In order to keep the rod moving at constant velocity, the force the hand exerts on the rod must balance the magnetic force on the current:

$$F_{hand} = F_m = ILB$$

Motional $\mathcal{E}$ mf

Motional $\mathcal{E}$ mf and Electrical Energy.



- Work done to maintain constant velocity:

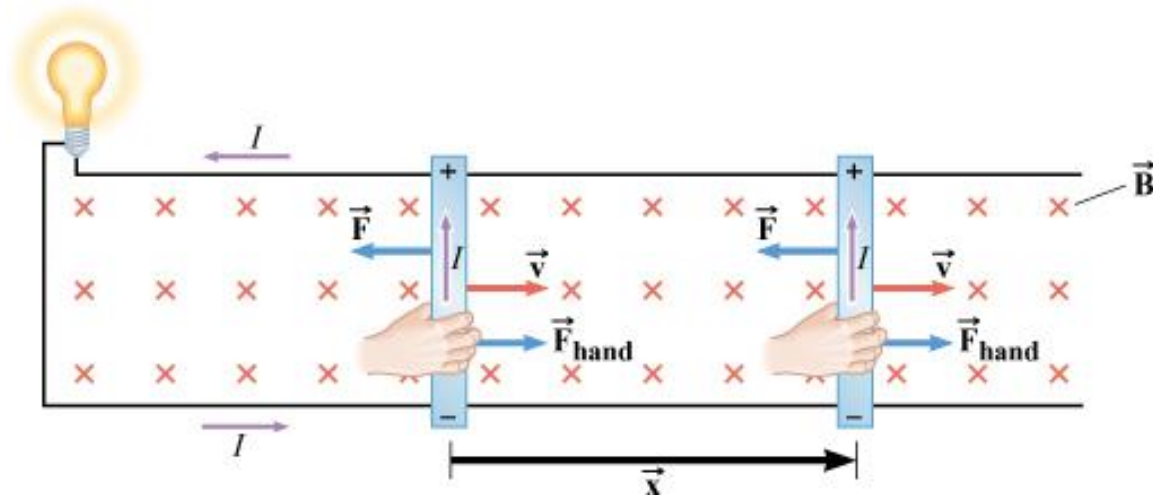
$$W = F_{hand} \cdot x = ILBx$$

- Rate of doing work (Power):

$$P = \frac{dW}{dt} = F_{hand} \cdot \frac{dx}{dt} = ILBv = I\mathcal{E}$$

Motional $\mathcal{E}$ mf

Motional $\mathcal{E}$ mf and Electrical Energy.



- Magnetic flux:

$$\Phi_B = BA = B(Lx)$$

- The induced $\mathcal{E}$ mf :

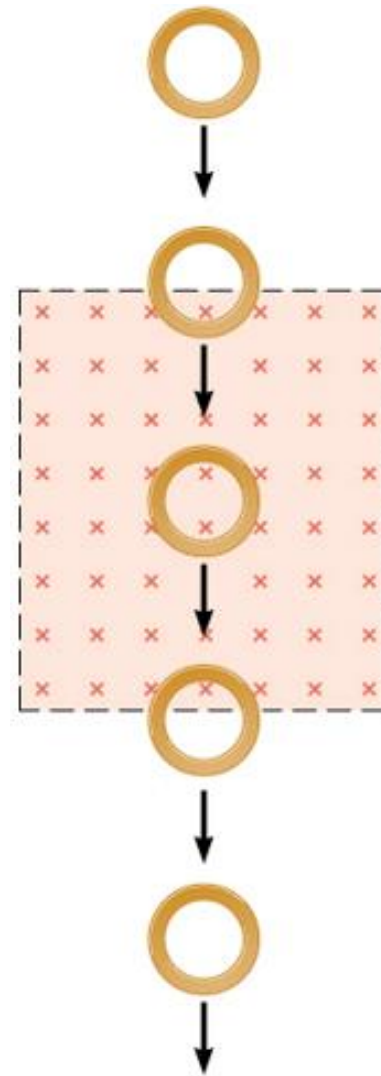
$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -BL\frac{dx}{dt} = -BLv$$

Motional $\mathcal{E}$ mf

Example: The Emf Produced by a Falling Copper Ring.

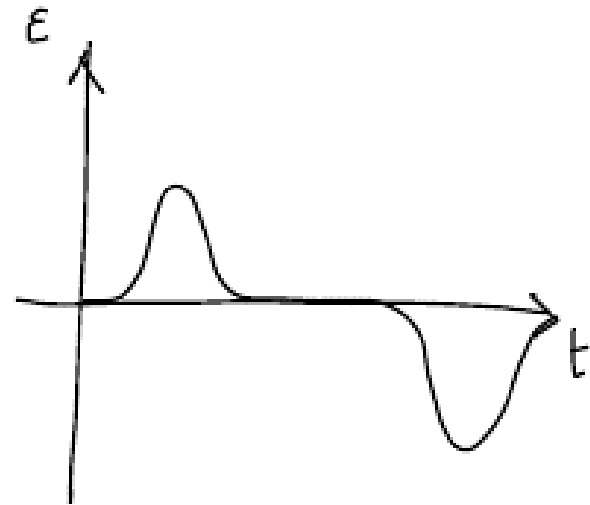
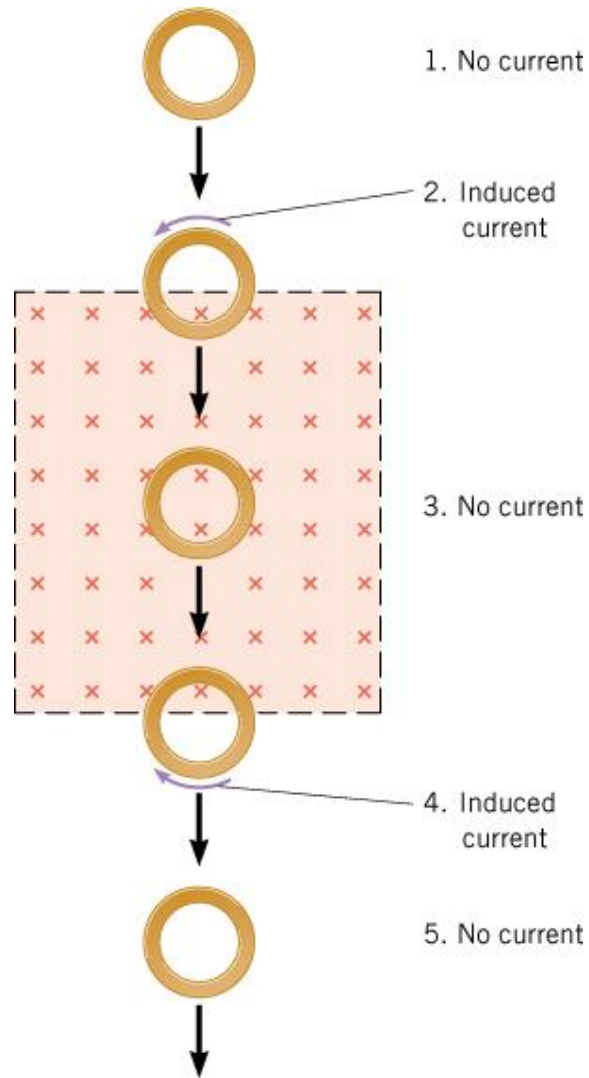
There is a constant magnetic field directed into the page in the shaded region. The field is zero outside the shaded region. A copper ring falls through the region.

For each of the five positions, determine whether an induced current exists and, if so, find its direction.



Motional $\mathcal{E}$ mf

Example: The Emf Produced by a Falling Copper Ring.



Induced Electric Field

An induced $\mathcal{E}$ mf occurs when there is a changing flux through a *stationary conductor*. What is it that pushes the charges around the circuit in this type of situation?

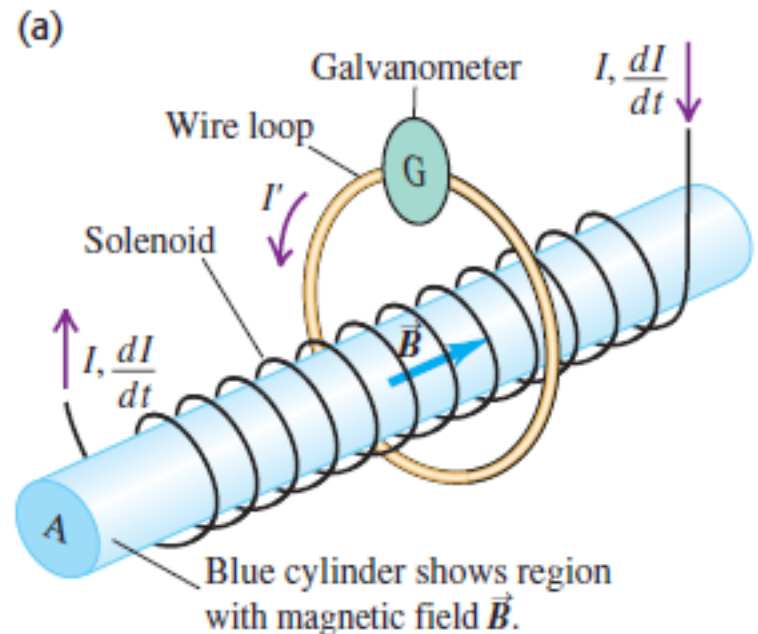
- Consider a long, thin solenoid with cross-sectional area A and n turns per unit length is encircled at its center by a circular conducting loop.

- Field of solenoid

$$B = \mu_0 n I$$

- Flux through the loop

$$\Phi_B = BA = (\mu_0 n I) A$$



Induced Electric Field

An induced $\mathcal{E}$ mf occurs when there is a changing flux through a *stationary conductor*. What is it that pushes the charges around the circuit in this type of situation?

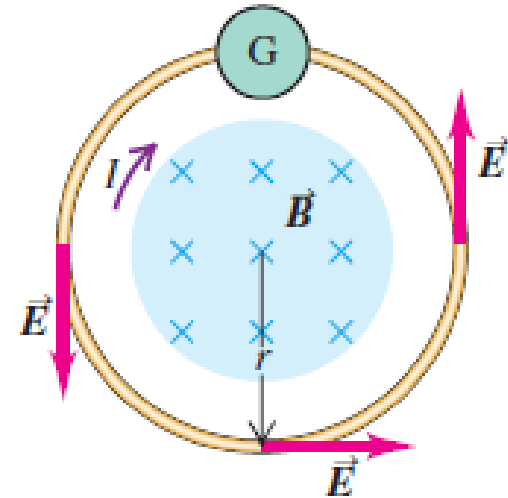
- When the solenoid current changes

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -(\mu_0 n A) \frac{dI}{dt}$$

- If the total resistance of the loop is R , the *induced current*

$$I' = \mathcal{E}/R$$

(b)



There has to be an **induced electric field** in the conductor
caused by the changing magnetic flux.

Induced Electric Field

Changing magnetic field acts as a source of *electric field*

- The **induced electric field** *does work on the charges*.

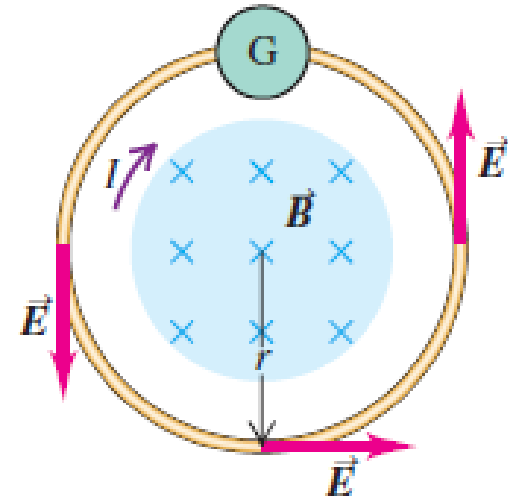
$$\oint \vec{E} \cdot d\vec{l} = \mathcal{E}$$

The electric field in the loop *is not conservative*

- Changing magnetic fields produce electric fields

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

(b)



END

